A Machine Learning–Driven SAEM Framework for Scalable Parameter Estimation in QSP

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Surrogate models efficiently replace large QSP models for faster **population parameter inference**

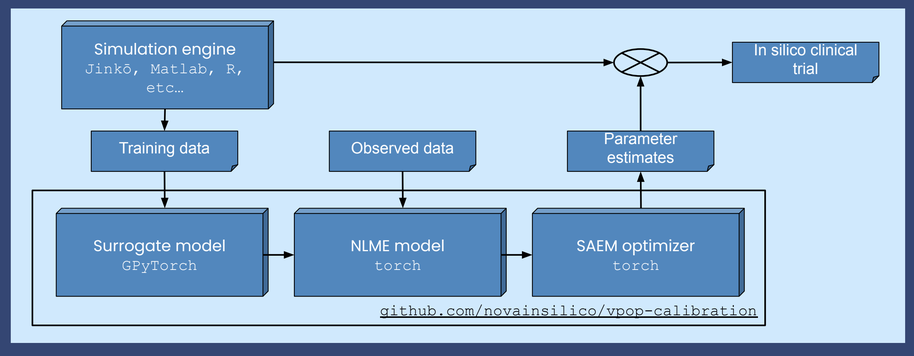

Figure 1: Surrogate modeling workflow for faster NLME optimization

Figure 2: Target-mediated drug disposition model from [5], used for synthetic data generation. Fixed parameter values were set as follows

```
k_off = 1.0 / h
k_D   = 0.5 nM
k_on  = k_off / k_D
k_deg = 0.5 / h
k_eP  = 0.2 / h
```

TMDD model equations

$$\frac{dL}{dt} = -k_{eL}L - k_{on}LR + k_{off}P,$$
$$\frac{dR}{dt} = k_{syn} - k_{deg}R - k_{on}LR + k_{off}P,$$
$$\frac{dP}{dt} = k_{on}LR + k_{off}P - k_{eP}P,$$
$$L(0) = 0, R(0) = R_0 = \frac{k_{syn}}{k_{deg}}, P(0) = 0,$$
$$DV = \log(\frac{L}{V_c})$$

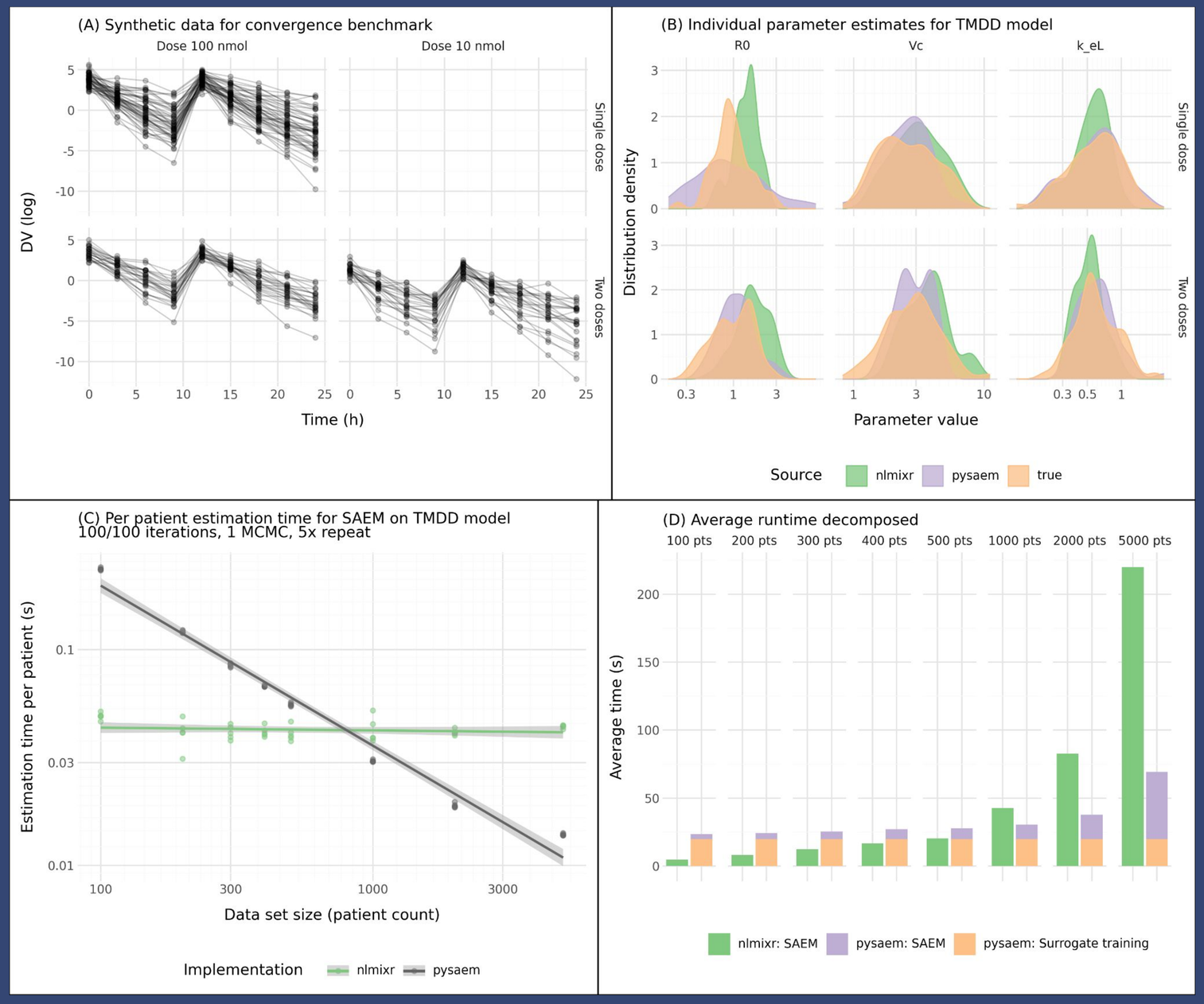

Figure 3: Parameter estimation benchmark between `nlmixr2` and `pysaem`

(A) Synthetic data for parameter estimation benchmark, generated on 50 patients with a simple TMDD model [5], and 3 parameters variability
- `R0 (nM)`: total available receptor concentration
- `Vc (L)`: volume of distribution
- `k_eL (1/h)`: antibody clearance rate

(B) Comparison of empirical bayesian estimators distributions for `nlmixr2` and `pysaem` estimations against ground truth, in a single dose and dual dose regimen

(C) Runtime scaling with observed data set size, in number of patients. Identical SAEM config: 100 iterations for exploration, 100 for smoothing phase, 1 Markov chain

(D) Runtime decomposition per algorithm: surrogate model training and SAEM estimation.

BACKGROUND

Determining optimal model granularity is a key challenge in model-informed drug development. Two approaches have gradually emerged, embodying the two facets of a modeling trade-off. Pharmacometrics on one hand, apply simple empirical models with low numbers of parameters, and leverage advanced statistical methods for population parameter inference, usually in pharmacokinetics (PK) models.
On the other hand, quantitative systems pharmacology (QSP) is a mechanistic modeling approach, providing a better interpretability of parameters and a reliable framework for translational studies or hypothesis testing. QSP models are often burdened with high dimensionality and computational complexity. This impedes the use of industry-standard methods for parameter inference from the pharmacometrics field.

METHODS

To bridge the gap between QSP and population calibration methods, we propose a surrogate model framework to enable the use of stochastic approximation expectation maximization (SAEM) [1] on large QSP models. Taking inspiration from the machine learning community [2], we present a Python library (powered by GPyTorch) for training surrogate Gaussian Processes (GP) on QSP model simulations, similarly to approaches developed by other research teams [3]. A PyTorch implementation of non-linear mixed effects (NLME) models is included, allowing for extremely efficient model prediction using surrogate models, as proposed in [4].
The optimization of NLME models using individual patient data is performed with a custom implementation of SAEM termed `pysaem`, enabling the estimation of population parameters and empirical bayesian estimators (EBE). An outline of the workflow and the tools is shown in Figure 1., and the full open-source code is available at github.com/novainsilico/vpop-calibration. All benchmarking examples cited or presented in this poster are available for reproducibility on the github repository.

RESULTS

The correctness of the SAEM implementation was first validated on typical examples from the literature that do not require an ODE model (orange trees, theophylline, results not shown). The method was then directly compared to nlmixr2 on a benchmarking study case, which requires the use of a stiff ODE model. A 3-equation model was used to generate synthetic observed data, and training data for the surrogate model, see Figure 2.

SAEM converges equally well when applied to a surrogate model

With a properly trained surrogate model, the `pysaem` implementation correctly estimated population parameters as well as individual EBEs. Figure 3 (A,B) shows a comparison of estimation results from both pysaem and nlmixr2 with identical control options (number of iterations, number of parallel MC chains, prior estimates).

Computational cost scaling of SAEM is drastically reduced with a surrogate

We then tested the computational cost scaling for this example study case with increasing sizes of observed data set. For large data sets, the surrogate model approach exhibits an improved scaling in time complexity, as evidenced by Figure 2 (C, D). For this toy model use case, the direct inference method remains faster for data sets of < 1k patients. For a larger QSP model however, if the per patient estimation cost is ~1 min, the total estimation time for 5k patients would reach ~80 hours, versus 80s for the surrogate approach.

CONCLUSION

The presented methods brings QSP models a step closer to estimation methods that allow for individual patient-level calibration on large data sets. The use of GPs as surrogates is a promising approach that unlocks the use of many other advanced numerical methods. In particular, Vpop generation via MCMC sampling could be accelerated as proposed by [3], but one could also use surrogate models for sensitivity analysis or real time model predictions for trial design optimization. This paves the way for novel applications of modeling in clinical pharmacology, applying complex models to large data sets.

REFERENCES

[1] Kuhn, Lavielle (2005) [2] Rasmussen, Willians (2006) [3] Myers et al. (2023) [4] Grenier et al. (2018) [5] Dua et al. (2015). Detailed references available upon request.